

A Theory of Economic Slack

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Preface

How much could incorporating slack into business cycle research really matter? Quite a lot. Let's go back to the start of the Great Recession in early 2008. The unemployment rate was 5.0%. This was actually below what modern macroeconomics stipulated. The unemployment target that comes out of the modern literature is the non-accelerating-inflation rate of unemployment (NAIRU): the unemployment rate that keeps inflation stable. Keeping inflation stable has nothing to do with the health of the labor market or the government's mandate of full employment, but because modern macroeconomics focuses on prices, the NAIRU has naturally become the preeminent unemployment target, including at the Federal Reserve.¹ The NAIRU was at 5.5% at the time (figure 1).

The unemployment rate skyrocketed as the Great Recession unfolded, reaching 9.9% by 2009. The NAIRU also rose to 6.6%. So according to the modern paradigm, the unemployment rate was 3.3pp too high at the peak of the recession. Furthermore, estimates of the NAIRU are surrounded by wide uncertainty, so it would not have been unreasonable to claim that the unemployment rate was only 2pp too high at the worst of the Great Recession.

By contrast, this book's theory of slack gives a full-employment rate of unemployment (FERU) below 4.0% during the recession (2008–2009), and below 4.5% at any point during the long recovery from the recession (2010–2015). This means that the unemployment rate was 5.9pp above the FERU in 2009, at the peak of the Great Recession (figure 1).²

¹According to the Council of Economic Advisers (2024, p. 24), “modern economics has generally defined full employment by citing the theoretical concept of the lowest unemployment rate consistent with stable inflation, which is referred to as u^* , ... the non-accelerating inflationary rate of unemployment (termed NAIRU).”

²Translating full employment into a number has long been a challenge for policymakers. The 1978 Full Employment and Balanced Growth Act did set an unemployment target of 3% within 5 years of the law, and 4% for young workers (Congress 1978). But these numbers were never taken seriously, in part because policymakers understood that the full-employment unemployment rate could not be a fixed number (DeLong 1996). When Congress asked Alan Greenspan in 1975 what the target for full employment should be, he quipped: “I do not think we should set a target” (Duboff 1977, p. 13).

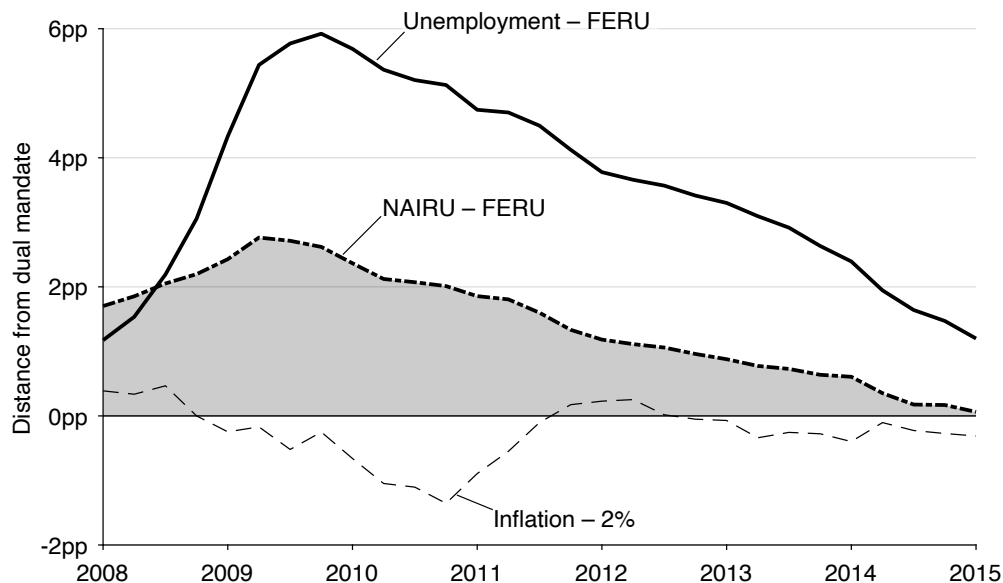


FIGURE 1. Costs of ignoring slack during the Great Recession in the United States

The unemployment rate is the quarterly average of the monthly, seasonally adjusted series produced by the Bureau of Labor Statistics (BLS 2025b). The inflation rate is the quarterly average of the monthly, seasonally adjusted, percentage change from a year ago of the consumer price index for all urban consumers, less food and energy, which is produced by the BLS (2025a). The non-accelerating-inflation rate of unemployment (NAIRU) is constructed by Crump et al. (2024, figure 2). The unemployment rates are reported as departures from the full-employment rate of unemployment (FERU), which is described in chapter 13. The inflation rate is reported as departure from the inflation target of 2%.

The picture painted by the FERU is therefore much more dire than what the NAIRU suggested. By ignoring slack, the modern paradigm set a target for unemployment that was much too lenient: 2.2pp too high in 2008, 2.8pp too high in 2009, and 2.4pp too high in 2010 (figure 1). It was not until 2015 that the gap between NAIRU and FERU vanished.

Assuming that policymakers would have tolerated the same unemployment gap under the FERU as they did under the NAIRU, how many jobs would have been saved? A lot, it turns out. On average between 2008 and 2015, the NAIRU was 1.5pp above the FERU, which translates into 16.3 million worker-years of unemployment.³ So, by neglecting slack during and after the Great Recession, the modern paradigm produced unemployment targets that were materially more permissive, allowing for an additional 16 million worker-years of unemployment. The focus on inflation embodied in the NAIRU is odd given that inflation remained 0.3pp below its target of 2% on average between 2008 and 2015 (figure 1).

Additionally, the FERU formula developed in this book would have helped dismiss

³Each quarter, we compute the gap between NAIRU and FERU and multiply it by the number of workers in the labor force. We then sum these values over the 32 quarters between 2008 and 2015. Dividing by 4 gives 16.3 million worker-years.

a hypothesis that was popular at the time of the Great Recession and hindered policy response. A substantial number of academic economists and policymakers argued that mismatch was a key cause of high unemployment and therefore that there was not much for the Federal Reserve to do—since it did not have the tool to tackle mismatch.⁴ One prominent advocate of that view was Narayana Kocherlakota (2010), who was President of the Federal Reserve Bank of Minneapolis at the time. He argued that the Beveridge curve broke down at the onset of the Great Recession, so that there was not much that the Fed could do about the elevated level of unemployment:

The relationship [between unemployment and job openings] has completely shattered.... Firms have jobs, but can't find appropriate workers. The workers want to work, but can't find appropriate jobs.... Whatever the source, though, it is hard to see how the Fed can do much to cure this problem.... Of course, the key question is: How much of the current unemployment rate is really due to mismatch, as opposed to conditions that the Fed can readily ameliorate? The answer seems to be a lot.... Most of the existing unemployment represents mismatch that is not readily amenable to monetary policy.

In essence, Kocherlakota argued that the shift in the Beveridge curve had drastically raised the unemployment rate the Fed should target. On this view, despite the high level of unemployment, the unemployment gap facing the Fed was not very large.

As we will see in the book, the Beveridge curve is indeed central to understanding the workings of the labor market, and is at the heart of our FERU formula. In turn, the FERU formula allows us to quantify how much the Fed's full-employment target increased during the Great Recession. It shows that mismatch only had a small effect on the FERU: yes, the Beveridge curve shifted outward; but no, the rise in the FERU did not account for most of the increase in unemployment. The FERU only increased from 3.8% at the beginning of 2008 to 4.3% in 2010 and 4.5% in 2012. Since the FERU did not increase much, but the unemployment rate increased drastically, the unemployment gap was substantial, peaking at 5.9pp at the end of 2009 (figure 1). This means that 5.9pp of unemployment reflected deficient demand and was amenable to monetary policy, and to fiscal policy as well. However, due to the confusion created by the shift in the Beveridge curve and the lack of theory to translate that shift into a new full-employment target, the policy response remained muted, contributing to the slow recovery from the recession.

So, who is this book for? I have aimed to write the book as a text in quantitative social science. As such, it requires some basic mathematical training, at roughly the undergraduate level, as well as a curiosity about the social sciences, since it draws at various points on evidence and theories from psychology, sociology, public health, and

⁴See Rothstein (2012) for an overview of the debate and related empirical evidence.

economic history. Anyone with this background should be able to read the book from beginning to end, absorb a coherent picture of how the economy fluctuates over the business cycle, and later apply the book’s theory to their own research.

Although some sections of the book are necessarily mathematical, I hope the ideas developed here will speak to general readers interested in how modern economies function and sometimes falter. I have aimed to make the data, models, and findings accessible to anyone who wishes to follow the central argument without every mathematical detail. To make this possible, the book relies throughout on theoretical diagrams and empirical plots so all arguments can be understood visually. Modern macroeconomics has abandoned models that can be represented and analyzed diagrammatically; but a theory that can be understood, manipulated, and extended through visual reasoning might have certain advantages over a theory that can only be understood through numerical solution of large dynamical systems.

It is true that this book diverges conceptually and methodologically from the traditional macroeconomic approach. Professional macroeconomists might find the material unfamiliar and difficult to engage with. When I presented the concept of FERU at a policy conference in Washington, DC, in 2023, Christina Romer criticized our effort to “divorce the concept of full employment from stable prices” on the grounds that “a concept of full employment that isn’t consistent with stable inflation is not a sensible goal for policy”.⁵ Romer is one of the most accomplished scholars of the Great Depression; an expert on business cycles; and a leading authority on monetary and stabilization policy.⁶ Furthermore, she understands how policy decisions are made in the real world: she chaired the Council of Economic Advisers at the time of the Great Recession, hired by the White House to guide the US economy out of the crisis. Her response reflected a common view—full employment should be defined through price stability—and revealed how inflation-centric current policy thinking is. Nevertheless, I hope that the book might entice macroeconomists to make a bit more room for slack in business cycle research.

⁵The quote is from the discussion that follows our paper on the FERU (Michaillat and Saez 2024, p. 422).

⁶See for instance Romer (1993), Romer (1999), and Romer and Romer (2023). Her expertise is widely recognized: she was codirector of the NBER Monetary Economics Program for two decades and is part of the NBER Business Cycle Dating Committee.

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